

Assignment TWO (Attempt all Questions individually)

- a. In a LCR series circuit derive the expression of Impedance, Phase difference between voltage and current, current in circuit, Wattless current, power factor, Power associated with circuit, Power at resonance, Resonant frequency and voltage at resonance, Quality factor and Bandwidth
- b. Graphically present variation of resistance, reactance and impedance with frequency
- c. Discuss application of LCR circuit
- d. A sinusoidal voltage of peak value 283 V and frequency 50 Hz is applied to a series LCR circuit in which $R = 3 \Omega$, $L = 25.48 \text{ mH}$, and $C = 796 \mu\text{F}$. Find
 - i. The impedance of the circuit
 - ii. The phase difference between the voltage across the source and the current
 - iii. The power dissipated in the circuit
 - iv. The power factor
- e. Suppose the frequency of the source can be varied, determine
 - i. The resonant frequency,
 - ii. The impedance,
 - iii. Resonant current and
 - iv. Power dissipated at the resonant conditions.